

How We See

Seeing is both a conscious and an unconscious action. Each type follows its own pathway in the brain. The conscious route helps recognize objects, while the unconscious route guides movement.

NEWBORN BABIES
CAN SEE ONLY
BLACK, WHITE,
AND RED



Cell area V1

Signals from the eyes are first received in the primary visual cortex (V1). Its neurons are sensitive to basic visual signals, including the orientation and direction of movement of objects and pattern recognition.



Cell area V2

In the secondary visual cortex (V2), some neurons improve on the images from V1, sharpening the lines and edges of complex shapes. Other neurons refine the initial interpretation of the color of objects.



Cell area V3

Visual area 3 (V3) is involved in analyzing angles, position, depth, and the orientation of shapes. It also helps process the direction and speed of objects. A few cells are also sensitive to color.

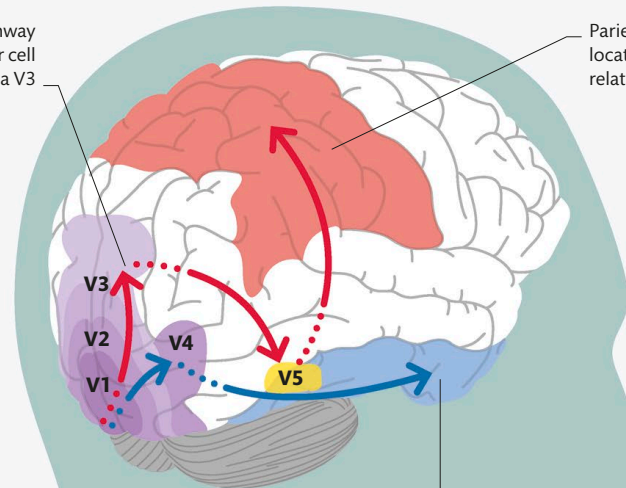
..... → **VISUAL CORTEX PATHWAY** →

Following the path

As visual information is processed by the layers of the visual cortex (see pp.68–69), it splits into two pathways known as the upper, or dorsal, route and the lower, or ventral, route. There is some uncertainty about where the split occurs, but the dorsal route handles our spatial awareness of where we are and how we move in relation to things around us, while the ventral route helps us identify, categorize, and recognize what we see. The dorsal route is important in assessing significant situations, particularly if instant action is required to avoid danger, such as moving away from a flying object. When this happens, the ventral route is relegated to a secondary position since the information it carries is not critical.

Visual pathway splits after cell area V3

Parietal lobes judge location of object in relation to observer



KEY

..... →
Dorsal route

..... →
Ventral route

Inferior temporal lobe involved in recognizing objects